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Subband Independent Component Analysis for Coherence Enhancement

Zhenghao Guo, [Yuhang Xu](#), Jan Rosenzweig, Verity McClelland, Ivana Rosenzweig, Zoran Cvetkovic

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Abstract

Objective: Cortico-muscular coherence (CMC) is becoming a common technique for detection and characterization of functional coupling between the motor cortex and muscle activity. It is typically evaluated between surface electromyogram (sEMG) and electroencephalogram (EEG) signals collected synchronously during controlled movement tasks. However, the

presence of noise and activities unrelated to observed motor tasks in sEMG and EEG results in low CMC levels, which often makes functional coupling difficult to detect. **Methods:** In this paper, we introduce Coherent Subband Independent Component Analysis (CoSICA) to enhance synchronous cortico-muscular components in mixtures captured by sEMG and EEG. The methodology relies on filter bank processing to decompose sEMG and EEG signals into frequency bands. Then, it applies independent component analysis along with a component selection algorithm for re-synthesis of sEMG and EEG designed to maximize CMC levels. **Results:** We demonstrate the effectiveness of the proposed method in increasing CMC levels across different signal-to-noise ratios first using simulated data. Using neurophysiological data, we then illustrate that CoSICA processing achieves a pronounced enhancement of original CMC. **Conclusion:** Our findings suggest that the proposed technique provides an effective framework for improving coherence detection. **Significance:** The proposed methodologies will eventually contribute to understanding of movement control and has high potential for translation into clinical practice.

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KW - filter banks

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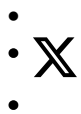
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